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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 4

Total Marks:30

Annexure A

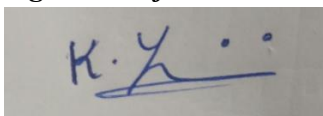
CASE STUDY

Code of Conduct:

I _K.Yamini(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature appears to be 'K. Yamini' with a stylized flourish at the end.

Annexure A CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1. $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2. $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

3. P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

4. Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF).

Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1. $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$

2. $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$

3. $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$

4. $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Total Marks (40)
Understanding of Case	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Application of Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Analysis	20% (2)	/2	/2	/2	/2	/8
Solution Design	20% (2)	/2	/2	/2	/2	/8
Clarity	10% (1)	/1	/1	/1	/1	/4
Total per Question	10 Marks	/10	/10	/10	/10	/40

Answers

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation (Understanding + Clarity)

Let:

F = Fever

C = Cough

R = Rash

B = Breathlessness

Flu = Patient has Flu

M = Patient has Measles

P = Patient has Pneumonia

Rules:

$$1. F \wedge C \rightarrow \text{Flu}$$

$$2. F \wedge R \rightarrow M$$

$$3. C \wedge B \rightarrow P$$

Facts:

F = True, C = True, B = True

(b) Forward Chaining (Application + Analysis)

Step 1: Check Rule 1

$$F \wedge C = \text{True} \wedge \text{True} = \text{True}$$

$\rightarrow$ Flu is derived

Step 2: Check Rule 3

$$C \wedge B = \text{True} \wedge \text{True} = \text{True}$$

$\rightarrow$ Pneumonia is derived

Step 3: Check Rule 2

$$F \wedge R \rightarrow M$$

R is False $\rightarrow$ Not applicable

Final Inference:

Flu = True

Pneumonia = True

(c) Backward Chaining (Goal: Pneumonia) (Solution Design)

Goal: P

Step 1: Find rule producing P

$C \wedge B \rightarrow P$

Step 2: Check sub-goals

C = True ✓

B = True ✓

Step 3: Both satisfied $\rightarrow$ Goal achieved

Conclusion: Patient A has Pneumonia

Case Study 2 — Autonomous Traffic Management

(a) Unification (Understanding + Application)

Rule:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Matching:

$x = \text{Ambulance}$

MGU (θ):

$\{ x / \text{Ambulance} \}$

Derived:

$\text{ClearPath}(\text{Ambulance})$

(b) Resolution Proof (Analysis + Application)

Convert to CNF:

1. $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
2. $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
3. $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts:

$\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

Negate Goal:

$\neg \text{Proceed}(\text{CarA})$

Resolution Steps:

Step 1:

From Rule1 $\rightarrow$ ClearPath(Ambulance)

Step 2:

From Rule2 $\rightarrow$ GreenSignal(Ambulance)

Step 3:

Apply Rule3 with substitution

$x = \text{CarA}, y = \text{Ambulance}$

$\text{Vehicle}(\text{CarA}) \wedge \text{Behind}(\text{CarA}, \text{Ambulance}) \wedge \text{GreenSignal}(\text{Ambulance})$

$\rightarrow \text{Proceed}(\text{CarA})$

Contradiction with $\neg \text{Proceed}(\text{CarA})$

Hence proved: $\text{Proceed}(\text{CarA})$

(c) Analysis of KB (Solution Design)

Problem:

- Only one emergency vehicle handled

Issue:

- Conflict when multiple emergencies

Required Rules:

- Priority handling rule
- Conflict resolution mechanism

Example:

$\text{Emergency}(x) \wedge \text{Emergency}(y) \wedge \text{Priority}(x) > \text{Priority}(y) \rightarrow \text{Serve}(x) \text{ first}$

Conclusion: Current KB is insufficient

Case Study 3 — Agricultural Expert System

(a) CNF Conversion (Understanding + Clarity)

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

$= \neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

$= \neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step1: Remove implication

$= \neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step2: Apply De Morgan

$= \text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts:

SoilDry, CropWheat

(b) Resolution Refutation (Analysis + Application)

Goal: ApplyDripMethod

Negate Goal:

$\neg$ ApplyDripMethod

Steps:

1. SoilDry $\rightarrow$ IrrigationNeeded
2. IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
3. Combine with $\neg$ ApplyDripMethod

Resolution:

ApplyDripMethod and $\neg$ ApplyDripMethod $\rightarrow \square$ (empty clause)

Hence proved: ApplyDripMethod is TRUE

(c) Removing SoilDry (Deep Analysis)

Without SoilDry:

Step 1:

IrrigationNeeded not derived

Step 2:

ApplyDripMethod not derived

Step 3:

Check P3:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Since ApplyDripMethod is False

$\rightarrow$ CropAtRisk = True

Conclusion: Crop becomes at risk

Case Study 4 — Wumpus World Agent

(a) Belief State (Understanding + Analysis)

Percepts:

Stench(1,2), Breeze(1,1), Glitter(2,2)

Using Modus Ponens:

1. Stench $\rightarrow$ Wumpus adjacent
 $\rightarrow$ Possible Wumpus: (1,1), (1,3), (2,2)

2. Breeze $\rightarrow$ Pit adjacent
 $\rightarrow$ Possible Pit: (1,2), (2,1)
3. Glitter $\rightarrow$ Gold(2,2)

Derived Knowledge:

- Gold at (2,2)
- Wumpus nearby
- Pit nearby

(b) Forward Chaining (Application)

Step 1:

Stench $\rightarrow$ Wumpus locations identified

Step 2:

Breeze $\rightarrow$ Pit locations identified

Step 3:

Combine knowledge:

- Dangerous cells identified

(c) Path Safety (Solution + Justification)

Path:

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

Analysis:

- [1,2] $\rightarrow$ Stench $\rightarrow$ Wumpus nearby
- [2,2] $\rightarrow$ Possible Wumpus location

Conclusion: Path is UNSAFE

Agent should avoid and choose alternative safe path